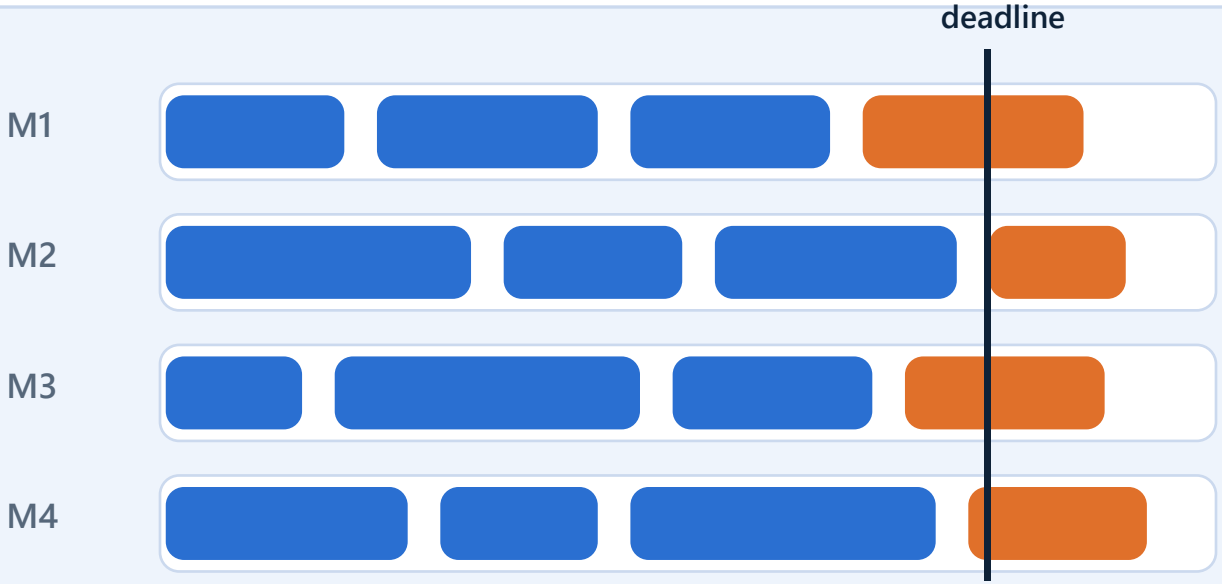
THE PROBLEM

Every cloud job is a decision: what runs next, and where?



A schedule: each block is a job on a machine. Orange jobs finish late.

94%

of enterprises already use cloud services [1]

$8,000 / min

average cost of cloud downtime (AUD) [1]

$1.7 trillion

expected global cloud market by 2033 (AUD) [1]

Same decision, many fields: data centres, hospitals, factories, deliveries.

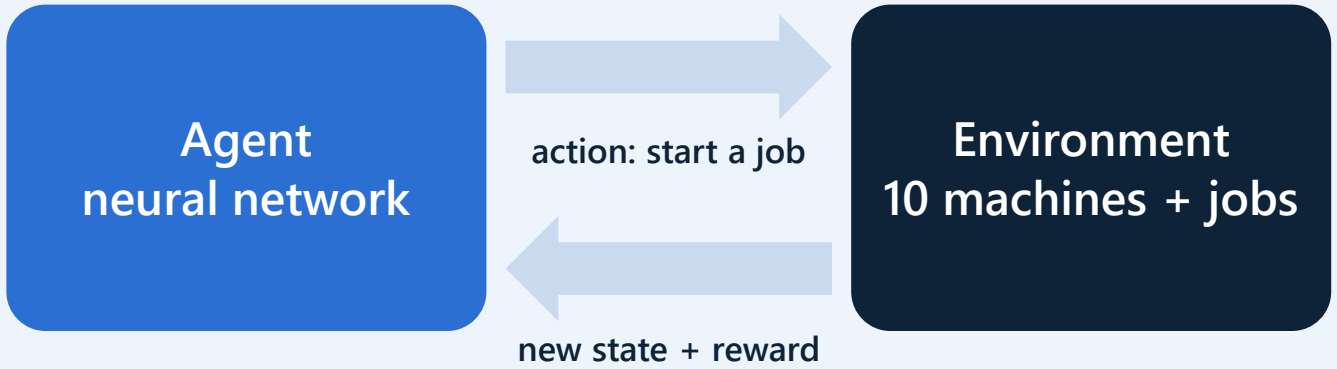
Our question: can reinforcement learning schedule better than the best hand-made rule?

[1] DigitalDefynd, "30 Surprising Cloud Computing Facts & Statistics," 2026.

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THE METHOD

The scheduler is a neural network that learns from reward



State

waiting jobs, deadlines, machine loads

Action

which job to start next (or wait)

Reward

minus the lateness cost

WHAT IT TRIES TO MINIMISE

$$J = \Sigma\, w \times (\text{lateness})^2$$

Late jobs hurt more the later they are; important jobs count more. Optional terms: late jobs, energy.

THREE KINDS OF SCHEDULER

Classic rules, made by hand
AI that learns on its own
AI that first copies a rule

Watch it learn

Train a tiny scheduler yourself and see its reward rise.

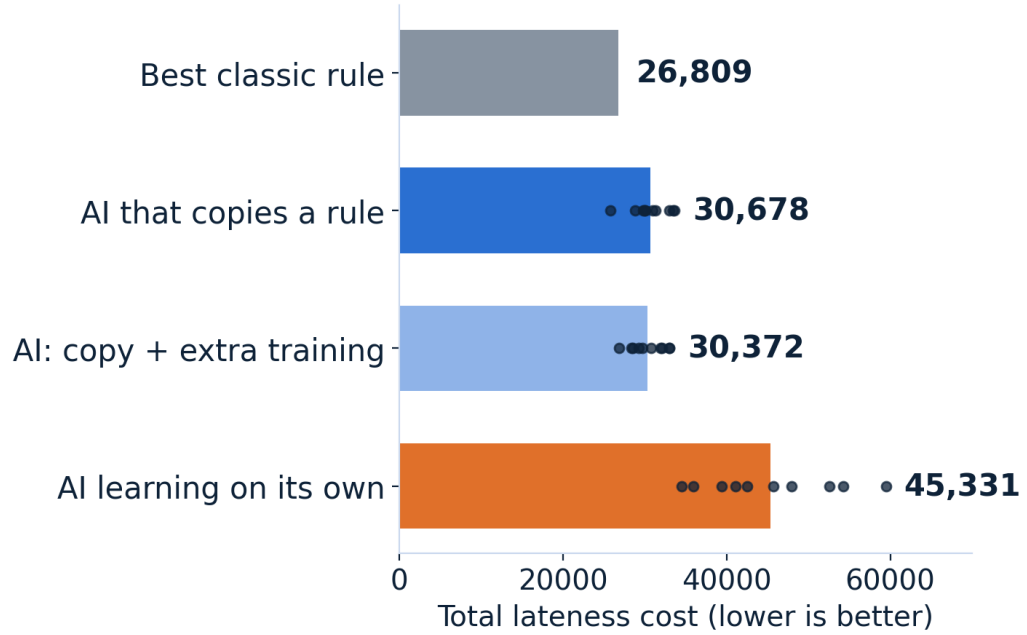
Interactive training demo →

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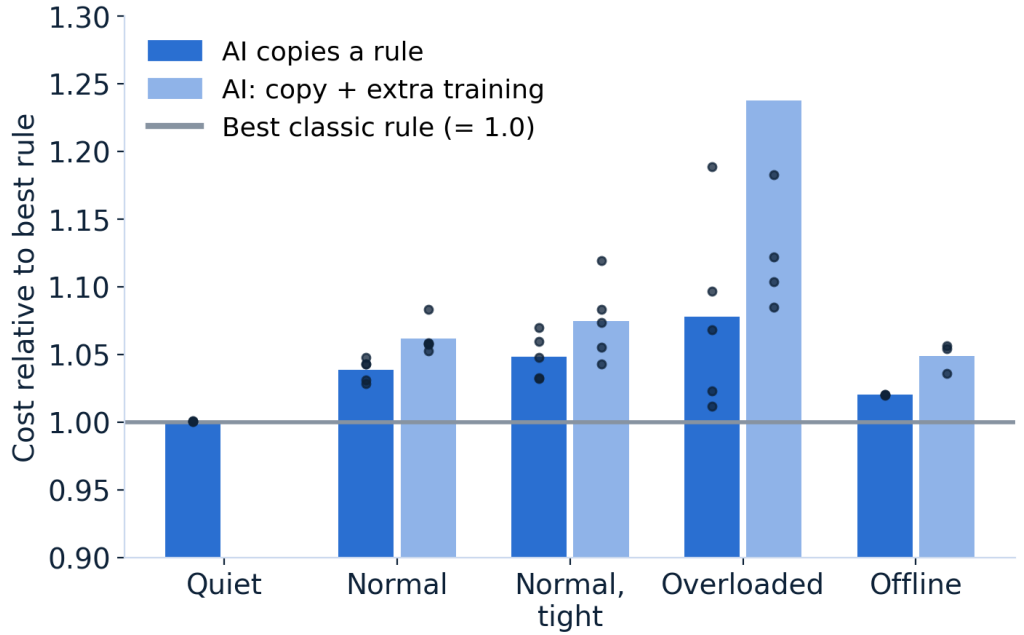
WHAT WE FOUND

The AI learned to match the best rules, but did not beat them

Busy machines: cost on 50 fresh test cases



Five scenarios: cost relative to the best rule



No AI scheduler was significantly better than the best classic rule.

Exploratory results: 50 fresh test cases per scenario, 3 to 10 training runs per bar (dots).

Race the models →

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From following a plan to leading the next project

Skills built

Modelling
a cloud problem turned into mathematics

Machine learning
reinforcement and imitation learning

Research rigour
fresh test data, honest reporting

Communication
interactive tools for non-experts

Where it leads

My direction
machine learning and mathematical optimisation

The thread
computation that makes hidden dynamics visible

Audience
from experts to curious visitors

What is next

Trade-offs
train for late jobs versus energy

Forecasts
can predicting arrivals help?

Understanding
why learning on its own is unstable

Leading
[my next step: own a project]

Year 2
CO_2 and quantum vibrations
MATLAB visualisation

Year 3
Cloud scheduling with RL
Python, interactive demos

Next
Lead a project
machine learning + optimisation

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